

My Reflective Journal: Chihuahua or Muffin?

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CNN Architecture

I used a Convolutional Neural Network (CNN) for this lab project and school assignment. I learned that a CNN is very good for pictures and clear image data. I think the CNN architecture has special layers inside the model. These are called convolutional layers and pooling layers in simple terms. I understand that these layers help the computer see shapes, small details, simple patterns, and basic forms, like the ears of a dog or the blueberries in a muffin.

I think this is very different from the traditional neural network I used in my previous workshop class at school. In my old network, I had to flatten all my pixels into one big line of raw number data. I think the old network lost a lot of important information about the shape and visual structure. My new CNN keeps the shape of the image in the correct original format, so I think it is much smarter and more useful.

Model Performance

I looked at my model performance after I finished training the system with many images. I am very happy because I got a good accuracy score on the final results. I got around 85% accuracy on my validation set during testing time. I looked at the pictures where my model made mistakes in some wrong predictions. I found some very interesting patterns in my misclassifications during error analysis. I saw that my model was very confused when a muffin had three dark blueberries in the middle center area. I think my model thought the blueberries were the two eyes and the nose of a Chihuahua dog face! I also saw that very fluffy Chihuahuas were sometimes classified as muffins in several test cases.

Comparison

I compared my new CNN model with my old traditional neural network model from last semester. I remember my old model had a very low accuracy for images in many test situations. I think my CNN is much better at performance in most practical cases. But I noticed a big difference in training time during the full process. I waited a very long time for my CNN to train on my computer system. I watched my computer screen for many minutes with slow progress updates. My traditional neural network was very fast and simple, but I

got bad results in the end. I think I prefer to wait longer so I can get a better model with higher accuracy.

Challenges and Solutions

I had many challenges when I did this lab project for my course. First, I had a problem with my file paths in the folder system. I could not load my data in Google Colab online platform environment. I was very frustrated and tired during that moment. I looked at my code and I searched on the internet for helpful answers. I solved my problem because I mounted my Google Drive correctly and I changed the directory path in my code file. Second, I had a challenge with overfitting in my training process. I saw my training accuracy was 99% but my validation accuracy was only 70% in comparison. I remembered my course materials and lecture notes from class. I used data augmentation as a smart solution method. I flipped my images and I rotated them in different simple angles. I trained my model again and I got better results with higher validation accuracy.

Real-World Applications

I thought about how I can use this in the real world outside school. I think I can use this type of image classification model in a factory work environment. If I work in a bakery factory, I can put a camera over the conveyor belt machine. I can use my model to find bad muffins or to sort different types of cakes quickly. I also think I can use this in a pet shelter animal center. If I receive hundreds of dog photos from different owners, I can use a model to automatically tag the dog breeds correctly. I think this can save a lot of time for workers in busy places.

Ethical Considerations

I also thought about ethical considerations in artificial intelligence systems. I read in my deep learning course materials that we must be careful with AI technology. I think one big problem is bias in training data. If I only train my model with pictures of brown Chihuahuas from one group, my model will not know what a white Chihuahua is in reality. I think this is unfair to the white dogs in general. In the real world, if I use a face camera for humans in public places, bias can be very bad and racist in serious ways. I must make sure I have a good dataset with many different pictures from diverse sources. I also think about privacy and personal data protection. If I collect pictures of dogs and muffins for research, I must

make sure I do not take pictures of people's faces without their clear permission. I want to build AI that is safe and good for everyone in society.

I really liked this lab experience in my course. I learned a lot of new concepts and I am proud of my CNN model and final results.